

User & Administrator Guide: Dynamic Command Configuration Engine (`axi_command_config`)

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Target Audience: Adept Developers, System Administrators, Technical QA Engineers

Applies To: Axi Command Palette (`Axi_Beta`) v0.4.0+ / Adept Web 11.4+

1. Overview & Purpose

The **Dynamic Command Configuration Engine** is a metadata-driven navigation architecture in the Axi Command Palette. Historically, adding or modifying command navigation routes required editing JavaScript source code (`axicmdmain.js`).

With the introduction of the `axi_command_config` table and REST API, **100% of command navigation is managed directly in the database**. System administrators and developers can register new commands, link them to forms, reports, or custom pages, and configure contextual parameters **without changing a single line of JavaScript**.

```
[User Types Command] → [Command Palette Autocomplete] → [Matcher: command + prompt_options]
→ [Query: axi_command_config] → [Placeholder Resolver: :username, :userroles, :appname]
→ [Execution Dispatcher: tstruct | iview | tstruct/iview | url | processflow | ivtoivload]
```

2. Database Schema Reference (`axi_command_config`)

The table `axi_command_config` is stored in the application database (PostgreSQL or Oracle).

Column Name	Data Type	Nullable	Description & Purpose
<code>config_id</code>	VARCHAR(50)	No	Primary Key. Unique identifier for the configuration rule (e.g. <code>cfg_configure_user</code>).
<code>command</code>	VARCHAR(50)	No	Root command verb (e.g. <code>Configure</code> , <code>SDK</code> , <code>Upload</code> , <code>Download</code> , <code>View</code>).
<code>prompt_options</code>	VARCHAR(200)	No	Sub-prompt option to match (e.g. <code>user</code> , <code>role listing</code> , <code>app</code> , <code>default</code>). Case-insensitive.
<code>prompt_id</code>	VARCHAR(50)	No	Target identifier: <code>TransID</code> for Tstructs, <code>IViewName</code> for Iviews, or filename for custom pages.
<code>prompt_option_type</code>	VARCHAR(20)	No	Execution mode determining how the client shell handles the navigation.
<code>param_field</code>	VARCHAR(100)	Yes	Query parameter key to bind the user's input parameter to (e.g. <code>name</code> , <code>type</code> , <code>actorname</code>).
<code>target_url</code>	VARCHAR(500)	Yes	Explicit target URL template override. If omitted, constructed dynamically based on <code>prompt_option_type</code> .
<code>extra_params</code>	VARCHAR(500)	Yes	Additional query string parameters or dynamic tokens to append (e.g. <code>&status=true&user=:username</code>).
<code>active</code>	VARCHAR(1)	Yes	Status flag (<code>'T'</code> for Active, <code>'F'</code> for Disabled). Default: <code>'T'</code> .

3. Supported Navigation Modes (`prompt_option_type`)

The engine provides specialized execution handlers tailored to Axpert structure types:

1. `tstruct` (Transaction Forms)

- **Use Case:** Opens an Axpert Tstruct form for creating a new record or editing an existing record.
- **Behavior:** If user provides a search value and `param_field` is specified, it opens in edit mode (`hltype=load`); otherwise opens in create mode (`hltype=open`).

```
INSERT INTO axi_command_config (config_id, command, prompt_options, prompt_id, prompt_option_type, param_field, hltype)
VALUES ('cfg_configure_user', 'Configure', 'user', 'axusr', 'tstruct', 'uname', 'T');
```

2. `iview` (Interactive Views / Reports)

- **Use Case:** Opens an Axpert interactive view report.
- **Behavior:** Routes directly to `iview.aspx?ivname=<prompt_id>`.

```
INSERT INTO axi_command_config (config_id, command, prompt_options, prompt_id, prompt_option_type, action, hltype)
VALUES ('cfg_configure_users', 'Configure', 'user listing', 'axusers', 'iview', 'T');
```

3. `tstruct/iview` or `iview/tstruct` (Dual-Mode Conditional Routing)

- **Use Case:** Single command that opens a listing report by default, but directly opens a specific record form if an entity name/ID is supplied.
- **Syntax for `prompt_id`:** `<TstructID>/<IviewName>` (e.g. `ad_ur/ad__url`).
- **Behavior:**
 - If user types `Configure Role` → Opens IView report `ad__url`.
 - If user types `Configure Role "Admin"` → Opens Tstruct form `ad_ur` with record `Admin`.

```
INSERT INTO axi_command_config (config_id, command, prompt_options, prompt_id, prompt_option_type, param_field, hltype)
VALUES ('cfg_role_dual', 'Configure', 'role', 'ad_ur/ad__url', 'tstruct/iview', 'rname', 'T');
```

4. `url` (Standard Custom ASPX / HTML Pages)

- **Use Case:** Redirects to custom Axpert web pages, tools, or configuration dashboards.
- **Behavior:** Resolves relative URLs to `../aspx/<prompt_id>` or uses explicit `target_url`.

```
INSERT INTO axi_command_config (config_id, command, prompt_options, prompt_id, prompt_option_type, param_field, hltype)
VALUES ('cfg_config_app', 'Configure', 'app', 'appconfig.aspx', 'url', NULL, '../aspx/appconfig.aspx');
```

5. `processflow` or `url/tstruct` (Process Builder Flow)

- **Use Case:** Opens visual workflow builder diagrams.
- **Behavior:** Routes to `processflow.aspx?loadcaption=AxProcessBuilder&processname=<paramValue>`. If no parameter is passed, falls back to `ad_pm` Tstruct.

```
INSERT INTO axi_command_config (config_id, command, prompt_options, prompt_id, prompt_option_type, par  
VALUES ('cfg_configure_peg', 'Configure', 'peg', 'ad_pm', 'processflow', 'processname', '../aspx/proce
```

6. `ivtoivload` (IView-to-IView Parameterized Navigation)

- **Use Case:** Navigates to parameterized summary Iviews with preset filter bindings.

```
INSERT INTO axi_command_config (config_id, command, prompt_options, prompt_id, prompt_option_type, par  
VALUES ('cfg_configure_dim_list', 'Configure', 'dimension listing', 'ad__upg', 'ivtoivload', 'prole',
```

4. Dynamic URL Placeholders

You can include contextual placeholders in `target_url` and `extra_params`. The client engine resolves these dynamically from the logged-in session before launching the target:

Placeholder Token	Resolved Runtime Value	Description
<code>:username</code>	<code>window.mainUserName</code>	The login username of the current user.
<code>:userroles</code>	<code>window.AxUserRoles</code>	Comma-separated list of active user roles.
<code>:userresp</code>	<code>window.userResp</code>	Active responsibility of the logged-in user.
<code>:appname</code>	<code>appname</code> / <code>window.mainProject</code>	Current application project schema name.
<code>:param</code>	<code>paramValue</code>	The parameter value typed or selected by the user.
<code>:paramField</code>	<code>paramField</code>	The field name configured for the target.

Placeholder Usage Example:

```
target_url: "../aspx/customdashboard.aspx?user=:username&role=:userroles&proj=:appname"
```

5. Step-by-Step Tutorial: Configuring a New Command

Scenario

You want to add a new command: **Configure SLA "Gold Policy"** that opens custom page `SlaConfig.aspx` with the selected policy name and current user's role.

Step 1: Insert the Configuration Row

```
INSERT INTO axi_command_config (
  config_id, command, prompt_options, prompt_id,
  prompt_option_type, param_field, target_url, extra_params, active
) VALUES (
  'cfg_configure_sla', 'Configure', 'sla', 'SlaConfig.aspx',
  'url', 'policyname', '../aspx/SlaConfig.aspx', 'role=:userroles&user=:username', 'T'
);
```

Step 2: Register Autocomplete Prompt (Optional)

```
UPDATE axi_command_prompts
SET prompt_values = prompt_values || ',SLA'
WHERE command_id = (SELECT command_id FROM axi_commands WHERE command_name = 'Configure')
AND prompt_position = 2;
```

Step 3: Test in the Command Palette

1. Open Axipt Web Shell and press `Ctrl + Space` to activate Axi Command Palette.
2. Type: `Configure SLA "Gold Policy"`
3. Press `Enter` or click **Go**.
4. The system executes: `../aspx/SlaConfig.aspx?policyname=Gold%20Policy&role=Developer&user=admin`
5. Click **Pop** to test tabbed execution in `PopupContainer.html`.

6. Troubleshooting & FAQs

- **Q: I added a row in `axi_command_config`, but the command palette says "Unknown command".**
A: The configuration is loaded on shell initialization. Refresh the browser (`Ctrl + F5`) or click the **Refresh** button on the Axi toolbar. Verify that `command` matches the root verb exactly and `prompt_options` is lowercase.
- **Q: How do I create a fallback route for an entire command group?**
A: Set `prompt_options` to `'default'`. If no exact sub-option matches, the engine falls back to this rule.
- **Q: My custom page opens in the main window, but I want it to open in a popup tab.**
A: Users can press `Ctrl + Shift + Enter` or select the **Pop** action button. The engine automatically appends `&AxisPop=true` and loads the page within `PopupContainer.html`.
- **Q: Can I temporarily disable a command without deleting the row?**
A: Yes. Update the status flag: `UPDATE axi_command_config SET active = 'F' WHERE config_id = 'cfg_configure_sla';`